Nature of Environments

by Prawin Raja

Specifying the task environment

- PEAS
- - Performance
- - Environment
- - Actuator
- - Sensors

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Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

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Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

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Fully Observable vs Partially Observable

Fully Observable:

- The agent has complete access to all the information about the environment at each point in time.
- An environment is fully observable if the agent's sensors give it access to the complete state of the environment at each point in time.
- Chess, where the positions of all pieces are visible to both players.
- Autonomous warehouse robots that can see all items and obstacles within their operational area through cameras and sensors.

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Fully Observable vs Partially Observable

Partially Observable

- The agent has incomplete or noisy information about the environment.
- An environment is partially observable if the agent's sensors do not give it access to the complete state of the environment.
- Poker, where players cannot see each other's cards.
- Self-driving cars that may have limited visibility due to fog, rain, or obstructions.

Single agent vs Multi agent

Single Agent

- An environment is single-agent if only one agent is operating in it.
- A puzzle game where only one player (agent) is involved.

Multi agent

- An environment is multi agent if multiple agents are operating in it.
- can be either ***competitive or cooperative***.
- Chess is competitive multi agent
- Taxi driving environment is cooperative trying to avoid collision
- In cooperative , proper communication is rational behavior
- In competitive , randomized behavior is rational to avoid predictability.

Deterministic vs Non-Deterministic

Deterministic

- if the next state is completely determined by the current state and the action executed by the agent.
- Tic-tac-toe, where the outcome of a move is predictable.
- Industrial robots performing assembly tasks with predefined sequences.

Deterministic vs Non-Deterministic

Non Deterministic (Stochastic)

- if the next state is not completely determined by the current state and the action executed by the agent.
- Most real situations are so complex that it is impossible to keep track of all the unobserved aspects; for practical purposes, they must be treated as non-deterministic.
- **Stochastic** - implies that uncertainty about outcomes is quantified in terms of *probabilities* (eg. 25% chance of rain tomorrow)
- a nondeterministic environment is one in which actions are characterized by their possible outcomes, but no probabilities are attached to them. (There's chance of rain tomorrow)

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Episodic vs Sequential

Episodic

- if the agent’s experience is divided into atomic episodes, where each episode consists of the agent perceiving and then acting.
- Spam email filtering where each email is assessed independently for spam characteristics.
- Robot that spot defective parts in an assembly line.

Episodic vs Sequential

Sequential

- if the current decision could affect all future decisions.
- A chess game where each move influences subsequent moves
- Its complex compared to episodic as it needs to think a lot.

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Static vs Dynamic

Static

- The environment does not change while the agent is making decisions.
- Solving a crossword puzzle where the puzzle remains the same throughout.

Static vs Dynamic

Dynamic

- if it can change while the agent is deliberating.
- Self-driving car navigating through traffic.
- are continuously asking the agent what it wants to do; if it hasn't decided yet, that counts as deciding to do nothing.

Static vs Dynamic

Semi-Dynamic

- is one where the environment remains static while the agent is deliberating, but the agent's performance measure or other relevant factors might change over time.
- In an online auction, the auction environment (the list of items) remains static, but the current highest bid can change over time as more bids are placed.
- Chess , when played with ticking clock.

Known vs Unknown

Known

- if the agent understands all aspects of the environment, including how actions affect the environment and the outcomes of all actions.
- Law of physics
- In manufacturing, robots operate in a highly controlled and predictable environment where all variables and actions are predefined.

Known vs Unknown

Unknown

- if the agent has no prior knowledge about the effects of its actions or the dynamics of the environment, and it must learn these through interaction.
- While driving in a new city, the car must navigate without prior detailed knowledge of local road conditions and must adapt to new situations.

Discrete vs Continuous

Discrete

- An environment is discrete if it has a finite number of distinct states and actions.
- A board game like chess with a finite number of positions and moves.

Discrete vs Continuous

Continuous

- An environment is continuous if it has an infinite number of states or actions.
- A robotic arm moving in a continuous 3D space.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

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